

Testing $f(Q)$ Gravity with Logarithmic Equation of State Using Latest Cosmological Data

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In this paper, we investigate the late-time accelerated expansion of the Universe in power law $f(Q)$ gravity, where a logarithmic dark energy parametrization is considered: $\omega_{de}(z) = \omega_0 + \omega_1 \ln(1+z)$. This description gives a smooth deviation from a constant equation of state within the complete range of redshifts. We obtain an analytical expression for the Hubble parameter and we use a Markov Chain Monte Carlo to compare the model with the most recent observational data obtained from Pantheon⁺ Type Ia supernovae, Baryon Acoustic Oscillation measurements from the second data release (DR2) of the Dark Energy Spectroscopic Instrument, and Cosmic Chronometers. We then perform a statistical comparison between our model and the standard Λ CDM model using the Akaike Information Criterion and the Bayesian Information Criterion. From our results, we conclude that the use of logarithmic parametrization in the $f(Q)$ gravity model $f(Q) = Q + 6\gamma H_0^2 \left(\frac{Q}{Q_0}\right)^n$ is a valid and more flexible alternative to standard dark energy models, as it provides a richer phenomenology at low redshifts.

Keywords: $f(Q)$ gravity; logarithmic equation of state; late-time cosmic acceleration; dark energy.

I. INTRODUCTION

Over the past decades, the observational discovery of the accelerated expansion of the Universe [1, 2] has profoundly challenged our understanding of gravitation and cosmic dynamics. Explaining this phenomenon remains one of the major challenges in modern cosmology. Two main theoretical approaches have been explored to account for the late-time accelerated expansion of the Universe: either introducing a new energy component, commonly referred to as dark energy (DE) or modifying Einstein's general relativity. In the standard framework, this dark energy is typically modeled as a cosmological constant, Λ , characterized by a negative pressure that produces a repulsive gravitational effect on cosmological scales. Together with dark matter, which is responsible for the formation of large-scale structures, this constant forms the cornerstone of the Λ CDM model, which remains the most widely supported scenario by current observations. However, despite its remarkable success, the Λ CDM model is not free from theoretical difficulties. It faces two major problems related to the nature and origin of the cosmological constant [3–

10], as well as growing tensions between different measurements of the current expansion rate of the Universe. These limitations have motivated the exploration of alternative avenues, including dynamic forms of dark energy or modified theories of gravity. The equation of state parameter $\omega(z)$, defined as the ratio of a fluid's pressure to its energy density, is widely used to characterize the dynamics of dark energy. The simplest way to explain cosmological observations is to consider dark energy as a cosmological constant, corresponding to a redshift independent equation of state parameter, $\omega = -1$. Some introduce non-standard forms of dark energy within the framework of general relativity, including quintessence [11, 12], phantom models [7, 13–18], k-essence [19, 20], and holographic dark energy [21–28]. Others focus on modifying the gravitational sector of Einstein's equations, giving rise to modified gravity theories [29–39], anisotropic cosmological models [40–42], or scenarios involving interactions between dark matter and dark energy [43–45]. These extensions seek to account for the cosmic acceleration and other unresolved phenomena by providing mechanisms alternative to conventional dark energy, while remaining compatible with current observations [46–53]. In classical general relativity, gravity is understood as a manifestation of the curvature of spacetime, a view rooted in the equivalence principle. Within this framework, the dynamics of gravitation are fully determined by the distribution of matter energy density encoded in the energy-momentum tensor, the Ricci tensor and the scalar curvature, which play

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a central role in describing the underlying Riemannian geometry, where both torsion and non-metricity vanish. Although this formulation has been remarkably successful in explaining gravitational phenomena especially within the Solar System it faces significant challenges at cosmological and galactic scales. The late-time accelerated expansion of the Universe and the dynamics typically attributed to dark matter suggest that Einstein's theory, in its standard form, may be incomplete when applied beyond local gravitational systems. These limitations have motivated the development of several extensions of general relativity, either by enriching the Einstein–Hilbert Lagrangian or by modifying its geometric foundations. Such approaches have led to numerous important extensions, including $f(R)$ gravity [54], $f(G)$ gravity [55], $f(P)$ gravity [56], and Horndeski scalar-tensor theories [57], etc. From a broader geometric perspective, however, curvature is not the only quantity capable of encoding gravitational effects: torsion and non-metricity also constitute fundamental attributes of a general metric affine geometry. Incorporating these structures naturally gives rise to the teleparallel $f(T)$ theories [58], based on torsion, and the symmetric teleparallel equivalent of general relativity (STEGR) described by $f(Q)$ theories [59], where the non-metricity scalar Q governs the gravitational interaction. These formalisms offer promising avenues for revisiting the gravitational sector and exploring alternatives to dark energy within a more general geometric framework [87]. Indeed, $f(Q)$ gravity has recently emerged as a promising alternative within the symmetric teleparallel framework. A key advantage of this theory is that it leads to second order field equations, avoiding the higher order instabilities often encountered in curvature based extensions. In addition, its geometric formulation provides a natural mechanism to account for the late-time acceleration of the Universe without invoking an explicit dark energy component, making it particularly suitable for cosmological applications. Besides, these kind of gravities distinguish from general relativity at perturbative levels particularly at the growth of structures.

In this work, we examine the late-time acceleration of the Universe within the framework of symmetric teleparallel gravity, adopting a power law form of $f(Q)$ originally proposed in Refs [60, 61]. In this geometrical setting, gravitation is described through the non-metricity scalar, Q , in a flat and curvature-free spacetime. We consider the specific model $f(Q) = Q + 6\gamma H_0^2 \left(\frac{Q}{Q_0}\right)^n$ [84] with $Q_0 = 6H_0^2$, from which we construct an effective dark-energy component emerging naturally from the modified Friedmann equations. To allow deviations from the standard cosmological constant scenario, we employ a logarithmic parameterization of the dark energy equation of state, $\omega_{\text{de}}(z) = \omega_0 + \omega_1 \ln(1+z)$, which provides a smooth departure from $\omega = -1$ across all redshifts [65].

We derive an analytical expression for the Hubble expansion rate and confront the model with current observations by combining Pantheon⁺ Type Ia supernovae (SNIa), Baryon Acoustic Oscillation (BAO) measurements from

the second data release (DR2) of the Dark Energy Spectroscopic Instrument (DESI), and Cosmic Chronometers (CC). A Markov Chain Monte Carlo analysis is performed to constrain the parameter space $(h, \Omega_{m0}, \omega_0, \omega_1, n)$, allowing us to assess the observational viability of the theory. We subsequently assess our model against the standard Λ CDM framework using the Akaike Information Criterion (AIC) [71] and the Bayesian Information Criterion (BIC) [72]. We further analyze several cosmological diagnostics including the deceleration, jerk, snap, dark matter and dark energy densities, $Om(z)$ diagnostic, and statefinder parameters to probe the dynamical behavior of the model. Our results show that the logarithmic parametrization within this power law $f(Q)$ framework provides a flexible alternative to standard dark energy scenarios, offering a richer low redshift phenomenology and the potential to mitigate existing cosmological tensions.

The manuscript is structured as follows. In Section II, we present the basic formalism of $f(Q)$ gravity and the assumptions underlying the construction of the specific cosmological model. In Section III, we present the dynamics of power law $f(Q)$ gravity with a logarithmic equation of state for dark energy. In Section IV, we describe the observational data and statistical methodology. Section V is devoted to the numerical results and discussion. In Section VI, we examine the evolution of key cosmological quantities such as the deceleration, jerk, and snap parameters, the dark matter and dark energy densities, as well as the $Om(z)$ and statefinder diagnostics. Finally, in Section VII, we present a summary of the main results and our concluding remarks.

II. OVERVIEW OF $f(Q)$ GRAVITY

The $f(Q)$ gravity represents an extension of the symmetric teleparallel formulation of general relativity, defined on a spacetime with vanishing curvature and torsion [62, 63]. In differential geometry, particularly within Weyl-Cartan geometry the symmetric metric tensor $g_{\mu\nu}$ is used to measure the length of vectors, while an asymmetric connection $\Sigma^\gamma_{\mu\nu}$ determines covariant derivatives and parallel transport. In this framework, the general affine connection can be naturally decomposed into three components: the Christoffel symbol $\Gamma^\gamma_{\mu\nu}$, corresponding to the Levi-Civita part, the contorsion tensor $C^\gamma_{\mu\nu}$ related to torsion, and the disformation tensor $L^\gamma_{\mu\nu}$ describing non-metricity [64]

$$\Sigma^\gamma_{\mu\nu} = \Gamma^\gamma_{\mu\nu} + C^\gamma_{\mu\nu} + L^\gamma_{\mu\nu}. \quad (2.1)$$

The Levi-Civita connection of the metric, $g_{\mu\nu}$, is defined in its usual form

$$\Gamma^\gamma_{\mu\nu} \equiv \frac{1}{2}g^{\gamma\sigma} \left(\frac{\partial g_{\sigma\nu}}{\partial x^\mu} + \frac{\partial g_{\sigma\mu}}{\partial x^\nu} - \frac{\partial g_{\mu\nu}}{\partial x^\sigma} \right). \quad (2.2)$$

The contorsion tensor $C^\gamma_{\mu\nu}$ can be expressed as

$$C^\gamma_{\mu\nu} \equiv \frac{1}{2}T^\gamma_{\mu\nu} + T_{(\mu}{}^\gamma{}_{\nu)}. \quad (2.3)$$

In Eq. (2.3), $T^\gamma_{\mu\nu} \equiv 2\Sigma^\gamma_{[\mu\nu]}$ represents the torsion tensor. Lastly, the disformation tensor $L^\gamma_{\mu\nu}$ is obtained from the non-metricity tensor $Q_{\gamma\mu\nu}$ as follows

$$L^\gamma_{\mu\nu} \equiv -\frac{1}{2}g^{\gamma\sigma}(Q_{\mu\sigma\nu} + Q_{\nu\sigma\mu} - Q_{\sigma\mu\nu}). \quad (2.4)$$

In the equation provided, the non-metricity tensor $Q_{\gamma\mu\nu}$ is defined as the covariant derivative of the metric tensor with respect to the Weyl-Cartan connection $\Sigma^\gamma_{\mu\nu}$, represented as $Q_{\gamma\mu\nu} = \nabla_\gamma g_{\mu\nu}$.

The connection under consideration is assumed to be devoid of torsion and curvature in the current background. It represents a simple coordinate transformation from the trivial connection mentioned in [82]. Therefore, for a connection that is both flat and free of torsion, it can be parameterized as follows

$$\Sigma^\gamma_{\mu\beta} = \frac{\partial x^\gamma}{\partial \xi^\rho} \partial_\mu \partial_\beta \xi^\rho. \quad (2.5)$$

where $\xi^\gamma = \xi^\gamma(x^\mu)$ is an invertible relation. It is always possible to select a coordinate system where the connection becomes zero. This condition is referred to as the coincident gauge and has been employed in many studies of STEGR [64]. The Symmetric teleparallel gravity provides a geometric description of gravity that is equivalent to General Relativity (STEGR) when using coincident gauge coordinates, i.e., $\Sigma^\gamma_{\mu\nu} = 0$ and $C^\gamma_{\mu\nu} = 0$. Therefore, from (2.1), we can infer that [85]

$$\Gamma^\gamma_{\mu\nu} = -L^\gamma_{\mu\nu}. \quad (2.6)$$

In $f(Q)$ gravity, the gravitational action is modified by replacing the Ricci scalar R in the Einstein-Hilbert action with a function $f(Q)$, where Q is the non-metricity scalar. This scalar encapsulates the geometric property of non-metricity, allowing for a new interpretation of gravitational interactions that can potentially address cosmological phenomena such as dark energy and the accelerated expansion of the Universe. The action for $f(Q)$ gravity is given by

$$S = \int \left[\frac{1}{2}f(Q) + \mathcal{L}_m \right] d^4x \sqrt{-g} \quad (2.7)$$

where $\mathcal{L}_m$ is the matter Lagrangian density, and g is the determinant of the metric tensor $g_{\mu\nu}$. We have set $8\pi G = 1$. To derive the field equations, we vary this action with respect to the metric $g_{\mu\nu}$. The non-metricity tensor $Q_{\gamma\mu\nu}$ is defined as

$$Q_{\gamma\mu\nu} = \nabla_\gamma g_{\mu\nu}, \quad (2.8)$$

$$Q_\gamma = Q_{\gamma}{}^\mu{}_\mu, \quad \tilde{Q}_\gamma = Q^\mu{}_{\gamma\mu},$$

here, ∇ denotes the covariant derivative. The superpotential tensor, also known as the non-metricity conjugate, can be expressed as

$$4P^\gamma_{\mu\nu} = -Q^\gamma_{\mu\nu} + 2Q_{(\mu}{}^\gamma{}_{\nu)} + Q^\gamma g_{\mu\nu} - \tilde{Q}^\gamma g_{\mu\nu} - \delta^\gamma_{(\mu} Q_{\nu)}, \quad (2.9)$$

where the trace of the non-metricity tensor is given by

$$Q = -Q_{\gamma\mu\nu} P^{\gamma\mu\nu}. \quad (2.10)$$

The variation of the action, Eq. (2.7), with respect to the metric leads to the field equations of $f(Q)$ gravity, which can be written as follows [88]

$$\frac{-2}{\sqrt{-g}} \nabla_\alpha (\sqrt{-g} f_Q P^\alpha_{\mu\nu}) - \frac{1}{2} g_{\mu\nu} f - f_Q (P_{\mu\alpha\beta} Q^\alpha_\nu - 2Q_{\alpha\beta\mu} P^\alpha_\nu) = \kappa T_{\mu\nu} \quad (2.11)$$

Where the matter energy-momentum tensor is defined as

$$T_{\mu\nu} = -\frac{2}{\sqrt{-g}} \frac{\delta(\sqrt{-g} \mathcal{L}_m)}{\delta g^{\mu\nu}} \quad (2.12)$$

By varying the action with respect to the affine connection, we obtain the following result [86]

$$\nabla^\mu \nabla^\nu (\sqrt{-g} f_Q P^\gamma_{\mu\nu}) = 0. \quad (2.13)$$

The field equations in $f(Q)$ gravity preserve the conservation of the energy-momentum tensor and reduce to Einstein's equations for $f(Q) = Q$.

To apply the $f(Q)$ gravity in the cosmology setup, we consider a spatially flat, homogeneous, and isotropic Universe described by the Friedmann-Lemaître-Robertson-Walker (FLRW) metric

$$ds^2 = -dt^2 + a^2(t) (dx^2 + dy^2 + dz^2), \quad (2.14)$$

where t denotes the cosmic time and $a(t)$ is the scale factor. The Hubble parameter is defined as

$$H(t) \equiv \frac{\dot{a}}{a}, \quad (2.15)$$

and the cosmological redshift is related to the scale factor through $1 + z = a^{-1}$.

Within the framework of symmetric teleparallel gravity, the non-metricity scalar associated with the FLRW geometry takes the simple form

$$Q = 6H^2. \quad (2.16)$$

This relation allows the gravitational sector to be fully expressed in terms of the Hubble parameter and its derivatives, significantly simplifying the cosmological equations.

Furthermore, to derive the corresponding Friedmann equations, we assume that the cosmic matter content can be modeled as a perfect fluid, whose energy-momentum tensor is given by

$$T_{\mu\nu} = (\rho + p)u_\mu u_\nu + pg_{\mu\nu}, \quad (2.17)$$

where ρ and p denote the total energy density and pressure, respectively, and u^μ is the four-velocity of the fluid satisfying $u^\mu u_\mu = -1$.

To describe deviations from general relativity, we write the gravitational Lagrangian in the form

$$f(Q) = Q + F(Q), \quad (2.18)$$

where the function $F(Q)$ encodes the modified gravity contributions. By substituting the FLRW metric into the field equations of $f(Q)$ gravity, one obtains the generalized Friedmann equations

$$3H^2 = \rho + \frac{F}{2} - QF_Q, \quad (2.19)$$

$$(2QF_{QQ} + F_Q + 1)\dot{H} + \frac{1}{4}(Q + 2QF_Q - F) = -2p, \quad (2.20)$$

where $F_Q = dF/dQ$ and $F_{QQ} = d^2F/dQ^2$.

The total energy density is decomposed as $\rho = \rho_m + \rho_r$, corresponding to pressureless matter and radiation, respectively. The total pressure is $p = p_m + p_r$, with the matter pressure being negligible. Each matter component satisfies the standard conservation equation

$$\dot{\rho} + 3H(1 + \omega)\rho = 0, \quad (2.21)$$

where ω is the equation of state parameter of the corresponding fluid.

Using the explicit expression of the non-metricity scalar, the following relations are obtained

$$Q = 6H^2, \quad (2.22)$$

These relations establish a direct link between the non-metricity scalar and the cosmological dynamics, and they play a key role in determining the effective energy density and pressure arising from the modified gravity sector.

III. LOGARITHMIC DARK ENERGY EQUATION OF STATE

In this section, we investigate the dynamical consequences of adopting a logarithmic equation of state (EoS) for the dark energy component within the framework of the power law $f(Q)$ gravity model. Unlike constant or linear parametrizations, we consider the logarithmic form [65]

$$\omega_{de}(z) = \omega_0 + \omega_1 \ln(1 + z). \quad (3.1)$$

For a spatially flat FLRW Universe, the modified Friedmann equations take the form

$$3H^2 = \rho_r + \rho_m + \rho_{de}, \quad (3.2)$$

$$2\dot{H} + 3H^2 = -\frac{1}{3}\rho_r - p_{de}. \quad (3.3)$$

where ρ_r and ρ_m represent the energy densities of radiation and matter, respectively. In contrast, ρ_{de} and p_{de} describe the effective dark energy density and pressure that originate from the geometric contribution of the $f(Q)$ model

$$\rho_{de} = \frac{F}{2} - QF_Q, \quad (3.4)$$

$$p_{de} = 2\dot{H}(2QF_{QQ} + F_Q) - \rho_{de}. \quad (3.5)$$

Assuming no interaction among matter, radiation and dark energy, the conservation equations are

$$\dot{\rho}_r + 4H\rho_r = 0, \quad (3.6)$$

$$\dot{\rho}_m + 3H\rho_m = 0, \quad (3.7)$$

$$\dot{\rho}_{de} + 3H(1 + \omega_{de})\rho_{de} = 0. \quad (3.8)$$

The dark energy equation of state ω_{de} can be expressed in terms of the function $F(Q)$ and its derivatives as

$$\omega_{de} = -1 + \frac{4\dot{H}(2QF_{QQ} + F_Q)}{F - 2QF_Q}. \quad (3.9)$$

From Eqs. (3.6) and (3.7), the evolution of pressureless matter and radiation follows immediately, yielding the usual scalings $\rho_m \propto a^{-3}(t)$ and $\rho_r \propto a^{-4}(t)$.

A. Power Law Model for $F(Q)$

Now, we consider a specific functional form of $F(Q)$, chosen for its analytical simplicity and its ability to capture deviations from General Relativity, in a power law form

$$F(Q) = 6\gamma H_0^2 \left(\frac{Q}{Q_0} \right)^n, \quad (3.10)$$

where H_0 , γ , n , and Q_0 are constants. This choice is motivated by the fact that Friedmann equations form a system of ordinary differential equations, for which power-law and exponential solutions can be obtained. Therefore, we adopt the power-law form in our study. Using this functional form of $f(Q)$, we first obtain

$$\rho_{de}(z) = 6\gamma H_0^2(1-2n) \left(\frac{H}{H_0}\right)^{2n}, \quad (3.11)$$

$$\dot{\rho}_{de} = 12n\gamma H_0^2(1-2n) \left(\frac{H}{H_0}\right)^{2n} \frac{\dot{H}}{H}. \quad (3.12)$$

By substituting Eqs. (3.11) and (3.12) into the dark energy conservation (3.8), we obtain the key relation, together with the parametrization of the dark energy equation of state parameter. Taking into account the relation between H and Q , we obtain

$$2n \frac{\dot{H}}{H} + 3H(1 + \omega_{de}) = 0. \quad (3.13)$$

Furthermore, using the relation between the redshift z and the scale factor, namely $a(t) = 1/(1+z)$, the derivative with respect to cosmic time can be rewritten in terms of the redshift variable as $\frac{d}{dt} = -(1+z)H(z)\frac{d}{dz}$. Hence, we rewrite Eq. (3.13) as a z differential equation

$$-n(1+z) \frac{dH^2}{dz} + 3[1 + \omega_{de}(z)] H^2 = 0. \quad (3.14)$$

Once the Hubble function is determined, the full background dynamics follow straightforwardly. Both matter and radiation preserve their standard scaling behaviours, while the geometrical dark energy density inherits the same logarithmic modulation present in the EoS. The resulting Friedmann equation provides the basis for the observational analysis conducted in the subsequent section.

For consistency with standard cosmological analyses, we adopt the normalization

$$H = 100 h \text{ km s}^{-1} \text{ Mpc}^{-1}. \quad (3.21)$$

In this framework, the parameter γ is treated as a derived quantity, calculated from the matter density Ω_{m0} and the power law index n via the relation

$$\gamma(\Omega_{m0}, n) = \frac{1 - \Omega_{m0} - \Omega_{r0}}{1 - 2n}, \quad (3.22)$$

where Ω_{r0} denotes the present day radiation density parameter, which is fixed to $\Omega_{r0} = 8.4 \times 10^{-5}$. This choice

B. Logarithmic EoS

By taking into account the logarithmic parametrization, Eq. (3.1), in Eq. (3.14), we obtain

$$-n(1+z) \frac{dH^2}{dz} + 3[1 + \omega_0 + \omega_1 \ln(1+z)] H^2 = 0 \quad (3.15)$$

i.e.

$$\frac{dH^2}{H^2} = \frac{3}{n} \frac{1 + \omega_0 + \omega_1 \ln(1+z)}{1+z} dz. \quad (3.16)$$

The general solution can be written as

$$H^2(z) = H_0^2 (1+z)^{\frac{3(1+\omega_0)}{n}} \exp\left[\frac{3\omega_1}{2n} \ln^2(1+z)\right], \quad (3.17)$$

where H_0 is the present day Hubble rate. Using Eq. (3.17) in Eq.(3.11), the dark energy density ρ_{de} can be expressed as

$$\rho_{de}(z) = 3\gamma(1-2n)H_0^2(1+z)^{3(1+\omega_0)} \exp\left[\frac{3\omega_1}{2} \ln^2(1+z)\right]. \quad (3.18)$$

The matter density, ρ_m , and the radiation density, ρ_r , can also be expressed in terms of the redshift z

$$\rho_m(z) = \rho_{m0}(1+z)^3, \quad \rho_r(z) = \rho_{r0}(1+z)^4, \quad (3.19)$$

Thus, the Friedmann equation, Eq. (3.2), becomes

$$\frac{H^2(z)}{H_0^2} = \Omega_{r0}(1+z)^4 + \Omega_{m0}(1+z)^3 + \gamma(1-2n)(1+z)^{3(1+\omega_0)} \exp\left(\frac{3\omega_1}{2} \ln^2(1+z)\right). \quad (3.20)$$

ensures that the model reduces to the correctly normalized Hubble rate at the present epoch.

IV. OBSERVATIONAL DATA AND STATISTICAL METHODOLOGY

A. Methodology

In this subsection, we briefly describe the Markov Chain Monte Carlo (MCMC) methodology employed to constrain the free parameters of the power law $f(Q)$ model. The MCMC technique is widely used in cosmology to efficiently explore high dimensional parameter spaces and to derive robust posterior probability distributions for cosmological parameters [66].

The fundamental idea of the MCMC approach is to generate a Markov chain that samples the parameter space according to a target probability distribution. Each point in

the chain is proposed based on the previous one through a transition probability governed by a chosen proposal distribution, ensuring convergence toward the posterior distribution [73, 74].

In our analysis, We constrain the model parameters using three independent observational datasets: the Pantheon⁺ Type Ia supernova compilation, consisting of 1701 data points [68]; the DESI DR2 BAO dataset, which includes 12 data points [89]; and the Cosmic Chronometer (CC) measurements, comprising 32 data points [83]. These datasets provide complementary constraints on the expansion history of the Universe.

The best-fit values of the model parameters are obtained by maximizing the likelihood function, which is assumed to have a Gaussian form. The joint analysis of multiple datasets allows us to derive tightly constrained posterior distributions for the parameters h , Ω_{m0} , and n , as well as for the effective dark energy equation of state parameters ω_0 and ω_1 .

$$\mathcal{L} \propto \exp\left(-\frac{\chi^2}{2}\right), \quad (4.1)$$

where χ^2 denotes the pseudo chi-square estimator [66]. For independent datasets, the total likelihood is given by the product of the individual likelihoods, which leads to a total chi-square

$$\chi_{\text{tot}}^2 = \chi_{\text{CC}}^2 + \chi_{\text{Pantheon}^+}^2 + \chi_{\text{DESI DR2}}^2 \quad (4.2)$$

The explicit construction of the χ^2 functions associated with each dataset is presented in the following subsections. The resulting MCMC chains are then used to derive the marginalized posterior distributions and confidence intervals for all model parameters.

B. Datasets

1. Cosmic Chronometers

We use a compilation of 32 cosmic chronometer (CC) measurements of the Hubble parameter $H(z)$ in the redshift range $0 < z \lesssim 2$. The CC method provides a direct and model-independent estimate of the expansion rate based on the differential age evolution of passively evolving galaxies, through the relation

$$H(z) = -\frac{1}{1+z} \frac{dz}{dt}. \quad (4.3)$$

We adopt the dataset of Moresco et al. [83], including the full covariance matrix to account for correlated systematic uncertainties¹, such as those arising from metallicity,

young stellar population contamination, and stellar population synthesis modeling. The χ^2 function for the cosmic chronometer data is defined as

$$\chi_{\text{CC}}^2 = \Delta \mathbf{H}^T \mathbf{C}^{-1} \Delta \mathbf{H}, \quad (4.4)$$

where $\Delta \mathbf{H} = H_{\text{th}}(z_i) - H_{\text{obs}}(z_i)$, with $H_{\text{obs}}(z_i)$ and $H_{\text{th}}(z_i)$ denoting the observed and theoretical values of the Hubble parameter at redshift z_i , respectively. The covariance matrix $\mathbf{C}$ includes both statistical and systematic contributions.

2. Pantheon⁺ dataset

We include the Pantheon⁺ compilation of Type Ia supernovae [68]. It comprises 1701 data points obtained from 1550 supernovae covering the redshift range $0.001 \leq z \leq 2.3$. The chi-square function associated with the supernova dataset is given by

$$\chi_{\text{Pantheon}^+}^2 = \vec{F}^T \cdot \mathbf{C}_{\text{Pantheon}^+}^{-1} \cdot \vec{F}, \quad (4.5)$$

where $\mathbf{C}_{\text{Pantheon}^+}$ denotes the covariance matrix derived from the Pantheon⁺ dataset, incorporating both statistical and systematic uncertainties. The parameter vector is defined as $\theta = (h, \Omega_{m0}, \omega_0, \omega_1, n, M)$, where M represents the absolute magnitude. In this expression,

$$\vec{F}_i = m_{B,i} - M - \mu_{\text{model}}, \quad (4.6)$$

where $m_{B,i}$ and μ_{model} represent the observed apparent magnitudes and the theoretical distance modulus, respectively. The predicted distance modulus, μ_{model} , is computed according to the chosen cosmological model as follows

$$\mu_{\text{model}}(z) = m - M = 5 \log_{10} D_L(z) + 25, \quad (4.7)$$

where D_L represents the luminosity distance, defined as

$$D_L(z) = (1+z) \int_0^z \frac{dz^*}{H(z^*)}, \quad (4.8)$$

where, c denotes the speed of light. Unlike the original Pantheon dataset, the Pantheon+ compilation breaks the degeneracy between the absolute magnitude M and the Hubble constant H_0 . This is achieved by expressing the vector $\vec{F}$ in Eq. (4.5) in terms of the distance modulus of Type Ia supernovae hosted in Cepheid galaxies. Consequently, an independent constraint on M is obtained, leading to the following expression

$$\vec{F}'_i = \begin{cases} m_{B,i} - M - \mu_i^{\text{Ceph}} & i \in \text{Cepheid hosts} \\ m_{B,i} - M - \mu_{\text{model}}(z_i) & \text{otherwise} \end{cases} \quad (4.9)$$

where, μ_i^{Ceph} denotes the distance modulus of the Cepheid host galaxy for the i -th Type Ia supernova, independently determined using Cepheid calibrators. Accordingly, Eq. (4.5) can be rewritten as

¹ The covariance matrix used in this work is publicly available at https://github.com/Ahmadmehrabi/Cosmic_chronometer_data.

Table I: Priors for the parameter space $h, \Omega_{m0}, M, \omega_0, \omega_1, n$.

Parameter	Prior
h	(0.6, 0.8)
Ω_{m0}	(0, 1)
M	(-20, -19)
ω_0	(-2, 0)
ω_1	(-2, 2)
n	(-2, 2)
r_d	(100, 200)

$$\chi_{\text{SN}}^2 = \vec{F}'^T \cdot \mathbf{C}_{\text{Pantheon}^+}^{-1} \cdot \vec{F}'. \quad (4.10)$$

3. DESI DR2 dataset

We use the DESI DR2 Baryon Acoustic Oscillation (BAO) dataset to constrain the power law $f(Q)$ model. This dataset includes several tracers, namely Bright Galaxy Survey (BGS) galaxies, Luminous Red Galaxies (LRGs), Emission Line Galaxies (ELGs), quasars (QSO), and the Lyman- α forest, covering the redshift range $0.1 \leq z \leq 4.2$ [89].

From these observations, we consider the comoving angular diameter distance $D_M(z)/r_d$ and the Hubble distance $D_H(z)/r_d$, defined as

$$D_M(z) \equiv \int_0^z \frac{c dz'}{H(z')}, \quad \text{and} \quad D_H(z) \equiv \frac{c}{H(z)}, \quad (4.11)$$

where r_d is the sound horizon at the drag epoch. Assuming standard early-Universe physics, r_d can be defined as

$$r_d \equiv r_s(z_d) = \int_{z_d}^{\infty} \frac{c_s(z')}{H(z')} dz', \quad (4.12)$$

where z_d denotes the redshift of the drag epoch and c_s is the sound speed of the baryon-photon fluid. We also include the volume averaged distance

$$D_V(z) \equiv [z D_M^2(z) D_H(z)]^{1/3}. \quad (4.13)$$

The DESI DR2 BAO distance measurements used in this work can be found in Table III of [79]. For these, we use the covariance matrices provided in the DESI DR2 release.

The chi-square for the DESI BAO dataset is defined as

$$\chi_{\text{DESI}}^2 = \Delta D^T \mathbf{C}^{-1} \Delta D, \quad (4.14)$$

where $\Delta D = D_{\text{obs}} - D_{\text{th}}$ denotes the difference between observed and theoretical BAO measurements, and $\mathbf{C}$ is the covariance matrix. For uncorrelated data, this reduces to

$$\chi_{\text{DESI}}^2 = \sum_i \left(\frac{D_i^{\text{obs}} - D_i^{\text{th}}}{\sigma_i} \right)^2. \quad (4.15)$$

In this work, we consider three dataset combinations: Pantheon⁺ only, Pantheon⁺ + DESI DR2, and Pantheon⁺ + DESI DR2 + CC. The adopted prior ranges are listed in Table I.

V. NUMERICAL RESULTS AND DISCUSSION

Table II summarizes the mean values and the corresponding 68% confidence intervals of the cosmological parameters obtained for Λ CDM and power law $f(Q)$ models using three dataset combinations: Pantheon⁺ only, Pantheon⁺+DESI DR2, and Pantheon⁺+DESI DR2+CC. Overall, both models lead to stable and well constrained parameter estimates, providing a consistent description of the observational data.

For the MCMC analysis of the combined datasets, we adopt the same priors as for the CC sample. The resulting 1σ and 2σ confidence contours are shown in Fig. 1.

For the Hubble parameter, the inferred values remain tightly constrained across all data combinations. Using the Pantheon⁺ dataset alone, we obtain $h = 0.735 \pm 0.010$ and $h = 0.733 \pm 0.010$ for the Λ CDM and power law models, respectively, in good agreement with the SH0ES measurement ($H_0 = 73 \pm 2 \text{ km s}^{-1} \text{ Mpc}^{-1}$). Including DESI DR2 data leads to consistent results, with $h = 0.736 \pm 0.011$ and $h = 0.734 \pm 0.011$ for the Λ CDM and power law models, respectively. Finally, using the combined Pantheon⁺ + DESI DR2 + CC dataset, we obtain $h = 0.735 \pm 0.011$ and $h = 0.732 \pm 0.010$ for the Λ CDM and power law models, respectively. The close agreement between the two models indicates that the modified gravity framework accurately reproduces the background expansion history.

The matter density parameter Ω_{m0} is constrained around 0.334 ± 0.019 and $0.235_{-0.081}^{+0.10}$ using Pantheon⁺ alone, 0.3070 ± 0.0097 and $0.242_{-0.028}^{+0.036}$ with Pantheon⁺ + DESI DR2, and 0.3075 ± 0.0099 and $0.286_{-0.015}^{+0.017}$ for the combined Pantheon⁺ + DESI DR2 + CC dataset, for the Λ CDM and power law models, respectively. This result confirms that the power law $f(Q)$ model remains consistent with the standard matter content required by the observed cosmic expansion.

In the power law $f(Q)$ scenario, the dark energy sector is described through a logarithmic parameterization of the equation of state, as seen in Eq. (3.1), which allows for a mild redshift dependence. The constraints reported in Table II indicate $w_0 = -0.87_{-0.13}^{+0.15}$ and $w_1 = 0.28_{-0.28}^{+0.23}$ from Pantheon⁺, $w_0 = -0.877 \pm 0.056$ and $w_1 = 0.35_{-0.12}^{+0.14}$ from Pantheon⁺+DESI DR2, and $w_0 = -0.898 \pm 0.050$ and $w_1 = 0.065 \pm 0.087$ from Pantheon⁺+DESI DR2+CC. These results show that the combination of datasets progressively improves the constraints on both w_0 and w_1 , with the tightest bounds obtained when all data sets are included.

The posterior distributions and the joint confidence contours shown in Figs. 1 and 2 further illustrate the impact of this parametrization. Fig. 1 presents the marginalized confidence levels at 1σ and 2σ for Pantheon⁺ alone, Pantheon⁺+DESI DR2, and Pantheon⁺+DESI DR2+CC. Using the joint analysis, Fig. 2 presents the confidence contours for both the Λ CDM model and the power law $f(Q)$ model. A clear correlation between ω_0 and ω_1 is observed, reflecting their combined role in shaping the redshift evolu-

tion of $\omega_{\text{de}}(z)$. The contours also demonstrate that the inclusion of multiple datasets significantly tightens the constraints, leading to well localized confidence regions for all parameters. In particular, the power law index and the coupling parameter are well constrained around their best-fit values, indicating that the additional degrees of freedom introduced by the model are effectively supported by the data.

A. Information criteria

Since the cosmological models considered in this work involve different numbers of free parameters, the minimum chi-square statistic χ^2_{min} alone is not sufficient to provide a reliable model comparison. To assess the relative performance of models based on their theoretical predictions and their consistency with observational data, we employ three commonly used statistical criteria: the reduced chi-square χ^2_{red} , the corrected Akaike information criterion (AIC_c) [71], and the Bayesian information criterion (BIC) [72]. These quantities are defined as

$$\chi^2_{\text{red}} = \frac{-2 \ln \mathcal{L}_{\text{max}}}{N_d - N_p}, \quad (5.1)$$

$$\text{AIC}_c = -2 \ln \mathcal{L}_{\text{max}} + 2N_p + \frac{2N_p(N_p + 1)}{N_d - N_p - 1}, \quad (5.2)$$

and

$$\text{BIC} = -2 \ln \mathcal{L}_{\text{max}} + N_p \ln(N_d), \quad (5.3)$$

where $\mathcal{L}_{\text{max}}$ denotes the maximum likelihood, N_d represents the total number of data points, and N_p is the number of free parameters of the model. The AIC and BIC values, together with their differences relative to the reference Λ CDM model, are reported in Table III for the considered datasets.

To compare the statistical performance of the models, we consider the differences of the information criteria with respect to the reference Λ CDM model, defined as $\Delta\text{AIC} = \text{AIC}_{\text{model}} - \text{AIC}_{\Lambda\text{CDM}}$ and $\Delta\text{BIC} = \text{BIC}_{\text{model}} - \text{BIC}_{\Lambda\text{CDM}}$. In this framework, negative values indicate that the considered model is statistically favored over Λ CDM, while positive values imply a preference for the reference model. Typically, $\Delta\text{AIC} \lesssim 2$ suggests that the two models are statistically indistinguishable, whereas larger values indicate decreasing support for the alternative model. For the BIC criterion, $\Delta\text{BIC} > 6$ is commonly interpreted as strong evidence against the model under consideration.

For the Pantheon⁺ + DESI DR2 dataset, the power law $f(Q)$ model gives $\Delta\text{AIC} = 0.91$, indicating that both models are statistically comparable, with a slight preference for Λ CDM. For the Pantheon⁺ + DESI DR2 + CC dataset, we find $\Delta\text{AIC} = 1.06$, which also suggests no significant preference between the two models according to the AIC criterion.

However, the BIC yields $\Delta\text{BIC} = 17.26$ and 17.44 for Pantheon⁺ + DESI DR2 and Pantheon⁺ + DESI DR2 + CC, respectively. These values indicate a strong penalty on the power law $f(Q)$ model due to its larger number of free parameters, thus clearly favoring the Λ CDM model.

The Pantheon⁺+DESI DR2 dataset shows comparable performance between the Λ CDM and power law $f(Q)$ models according to the AIC criterion, while the addition of CC data further tightens the constraints. Taken together, these results demonstrate that the power law $f(Q)$ model provides a statistically consistent and phenomenologically viable description of the observational data. The logarithmic dark energy parameterization allows for a controlled departure from a strictly constant equation of state, while remaining closely aligned with the Λ CDM scenario within current observational constraints.

VI. COSMOGRAPHIC AND DYNAMICAL DIAGNOSTICS

Cosmography provides a model-independent framework to describe the expansion history of the Universe, relying only on the assumptions of homogeneity and isotropy. This approach allows the kinematical behavior of cosmic expansion to be characterized without specifying a particular gravitational theory or dark energy model, making it a valuable tool for confronting theoretical predictions with observational data.

The aim of this section is to investigate a cosmological model capable of describing the global dynamics of the Universe through its large-scale physical and geometrical properties. It is convenient to expand the scale factor $a(t)$ as a Taylor series around the present cosmic time t_0

$$\begin{aligned} \frac{a(t)}{a_0} = & 1 + \frac{(t - t_0)}{1!} \frac{da}{dt} \Big|_{t_0} + \frac{(t - t_0)^2}{2!} \frac{d^2a}{dt^2} \Big|_{t_0} \\ & + \frac{(t - t_0)^3}{3!} \frac{d^3a}{dt^3} + \frac{(t - t_0)^4}{4!} \frac{d^4a}{dt^4} + \dots \end{aligned} \quad (6.1)$$

where a_0 denotes the present value of the scale factor.

The coefficients appearing in this expansion can be expressed in terms of cosmographic parameters defined as

$$\begin{aligned} H &= \frac{1}{a} \frac{da}{dt}, & q(t) &= -\frac{1}{aH^2} \frac{d^2a}{dt^2}, \\ j(t) &= \frac{1}{aH^3} \frac{d^3a}{dt^3}, & s(t) &= \frac{1}{aH^4} \frac{d^4a}{dt^4}, \end{aligned} \quad (6.2)$$

which are commonly known as the Hubble, deceleration, jerk, and snap parameters. These quantities provide a purely kinematical description of the cosmic expansion and are sufficient to characterize the overall behavior of the Universe.

For practical purposes, it is useful to rewrite these parameters in terms of the redshift z [77, 78]. They can then

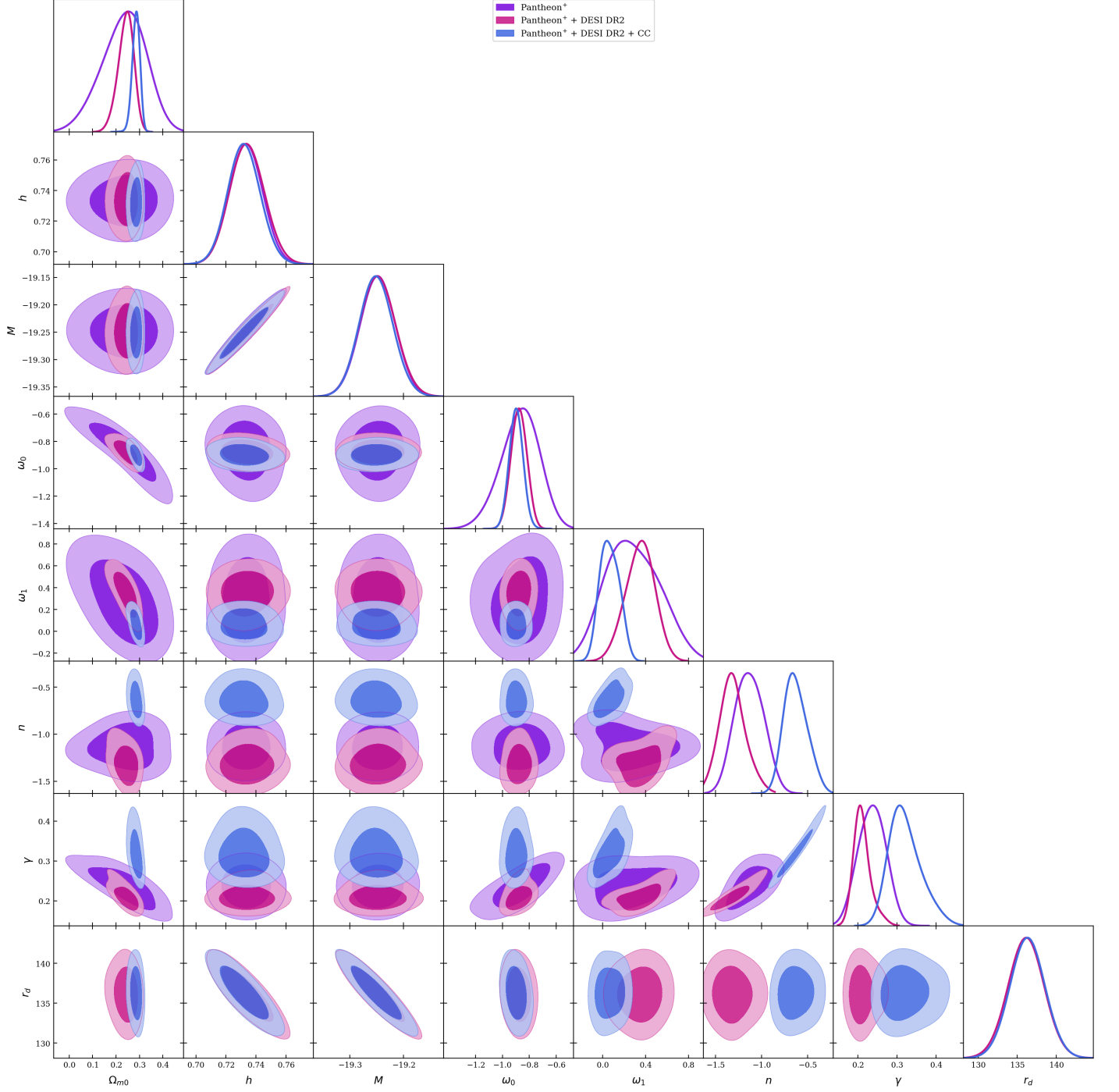


Figure 1: The 1σ and 2σ confidence contours and posterior distributions for the power law $f(Q)$ model using Pantheon⁺, DESI DR2, and CC datasets.

be expressed as

$$\begin{aligned}
 q(z) &= -1 + (1+z) \frac{E'(z)}{E(z)}, \\
 j(z) &= (1+z)^2 \frac{E''(z)}{E(z)} + q^2(z), \\
 s(z) &= -(1+z)j'(z) - 2j(z) - 3q(z)j(z),
 \end{aligned} \tag{6.3}$$

where $E(z) = \frac{H(z)}{H_0}$ denotes the normalized Hubble parameter.

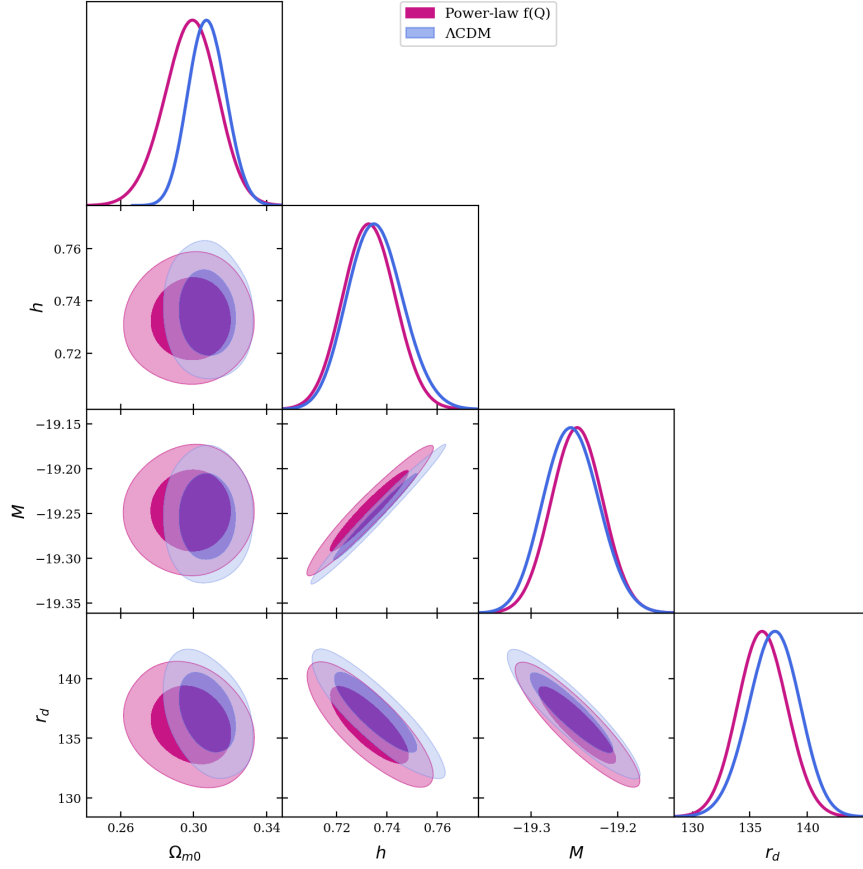


Figure 2: The 1σ and 2σ confidence contours, along with the posterior distributions, obtained for the power law $f(Q)$ and Λ CDM models using the Pantheon⁺ + DESI DR2 + CC datasets.

Table II: Mean values and standard deviations of the cosmological parameters for the Λ CDM and power law $f(Q)$ models.

Model	h	Ω_{m0}	M	ω_0	ω_1	n	γ	r_d
Pantheon ⁺								
Λ CDM	0.735 ± 0.010	0.334 ± 0.019	-19.246 ± 0.030	—	—	—	—	—
Power law	0.733 ± 0.010	$0.235^{+0.10}_{-0.081}$	-19.248 ± 0.030	$-0.87^{+0.15}_{-0.13}$	$0.28^{+0.23}_{-0.28}$	-1.13 ± 0.15	0.236 ± 0.033	—
Pantheon ⁺ + DESI DR2								
Λ CDM	0.736 ± 0.011	0.3070 ± 0.0097	-19.252 ± 0.031	—	—	—	—	137.0 ± 2.1
Power law	0.734 ± 0.011	$0.242^{+0.036}_{-0.028}$	-19.248 ± 0.031	-0.877 ± 0.056	$0.35^{+0.14}_{-0.12}$	$-1.32^{+0.12}_{-0.15}$	$0.210^{+0.015}_{-0.023}$	136.1 ± 2.2
Pantheon ⁺ + DESI DR2 + CC								
Λ CDM	0.735 ± 0.011	0.3075 ± 0.0099	-19.253 ± 0.031	—	—	—	—	137.1 ± 2.1
Power law	0.732 ± 0.010	$0.286^{+0.017}_{-0.015}$	-19.251 ± 0.030	-0.898 ± 0.050	0.065 ± 0.087	$-0.63^{+0.11}_{-0.14}$	$0.320^{+0.033}_{-0.047}$	136.3 ± 2.2

Table III: The corresponding $\chi^2_{\min}$, AICc and BIC for the examined cosmological models.

Model	$\chi^2_{\min}$	χ^2_{red}	AIC	Δ AIC	BIC	Δ BIC
Pantheon ⁺ + DESI DR2						
Λ CDM	1536.31	0.899	1544.31	0	1566.11	0
Power law	1531.22	0.897	1545.22	0.91	1583.37	17.26
Pantheon ⁺ + DESI DR2 + CC						
Λ CDM	1546.71	0.888	1554.71	0	1576.55	0
Power law	1541.77	0.887	1555.77	1.06	1593.99	17.44

A. Deceleration parameter

The deceleration parameter, shown in Fig. 3, exhibits a smooth transition from an early decelerated phase, as-

sociated with matter domination, to a late-time accelerated expansion. A positive value of q corresponds to a

decelerating Universe, while a negative value indicates an accelerating expansion. This behavior reflects the transition from deceleration to acceleration in the recent past, with transition redshifts $z_t \approx 0.635, 0.666, 0.679$ for the Pantheon⁺, Pantheon⁺+DES DR2, and Pantheon⁺+DES DR2+CC datasets, respectively. At the present epoch, $q_0 \approx -0.445, -0.455, -0.457$ for the same datasets, confirming the current accelerated expansion. The close agreement among these values underscores the robustness of the model with respect to the inclusion of additional data, while future predictions are sensitive to the data.

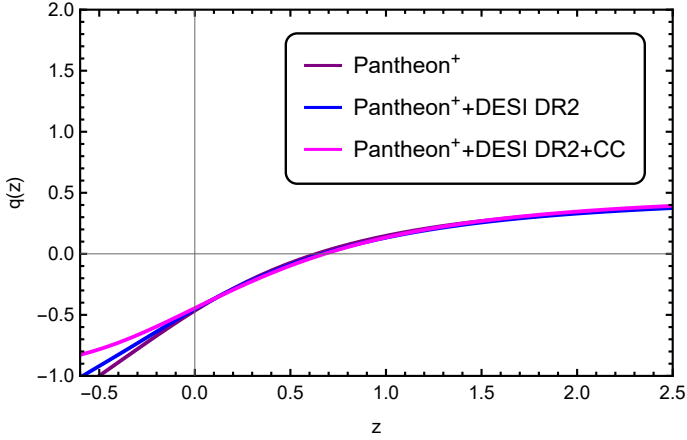


Figure 3: Evolution of the deceleration parameter as a function of redshift z .

B. Jerk parameter

The jerk parameter, shown in Fig. 4, evolves smoothly over the entire redshift range, with no indication of pathological behavior. At the present epoch, the values $j_0 \approx 0.81, 0.74, 0.70$ for the Pantheon⁺, Pantheon⁺+DES DR2, and Pantheon⁺+DES DR2+CC datasets, respectively, remain close to the Λ CDM reference value $j = 1$. The slight decrease of j_0 with the inclusion of additional datasets reflects tighter observational constraints, while preserving values near unity. This behavior indicates that the model reproduces the main kinematic features of the standard cosmological scenario with only mild deviations at late times. In the recent past and at earlier epochs, the evolution is not sensitive to the data, whereas future predictions become more sensitive to the choice and precision of the observational datasets.

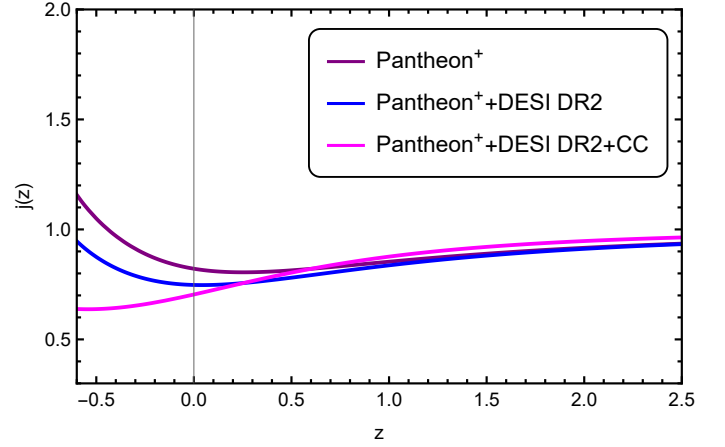


Figure 4: Evolution of the jerk parameter j as a function of redshift z .

C. Snap parameter

The evolution of the snap parameter $s(z)$ for the present model is illustrated in Fig. 5. This parameter captures higher order variations of the cosmic expansion and evolves in a smooth and regular manner across the full redshift range, indicating the stability of the model at higher derivatives. At the present epoch, s_0 takes the values -0.37 for the Pantheon⁺ dataset, -0.46 for Pantheon⁺+DES DR2, and -0.65 when CC data are included, highlighting the impact of additional observational constraints on higher order aspects of the expansion history. Nevertheless, its overall behavior remains well controlled and consistent with a smooth late-time expansion, confirming the robustness and reliability of the model in describing the recent evolution of the Universe. In the recent past and at earlier epochs, the evolution remains insensitive to the observational data, while future predictions become increasingly sensitive to the quality and precision of the datasets.

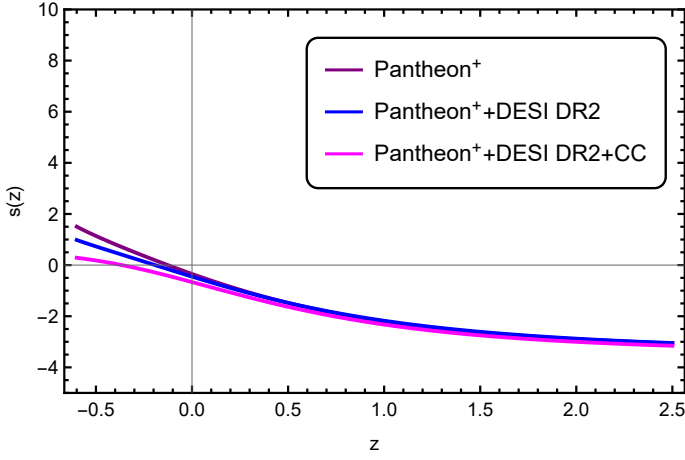


Figure 5: Evolution of the snap parameter s as a function of redshift z .

D. EoS Parameter

The evolution of the dark energy equation of state $\omega_{de}(z)$, shown in Fig. 6, exhibits a smooth and mild redshift dependence. At the present epoch ($z = 0$), the values are $\omega_{de} = -0.897$ for Pantheon⁺, $\omega_{de} = -0.878$ for Pantheon⁺+DESI DR2, and $\omega_{de} = -0.898$ for Pantheon⁺+DESI DR2+CC, indicating a quintessence-like behavior.

A crossing of the phantom divide line ($\omega_{de} = -1$) occurs at negative redshift, with $z \simeq -0.604$ for Pantheon⁺, $z \simeq -0.756$ for Pantheon⁺+DESI DR2, and $z \simeq -0.748$ for Pantheon⁺+DESI DR2+CC. This indicates that the transition to the phantom regime is shifted further into the future when additional datasets are included.

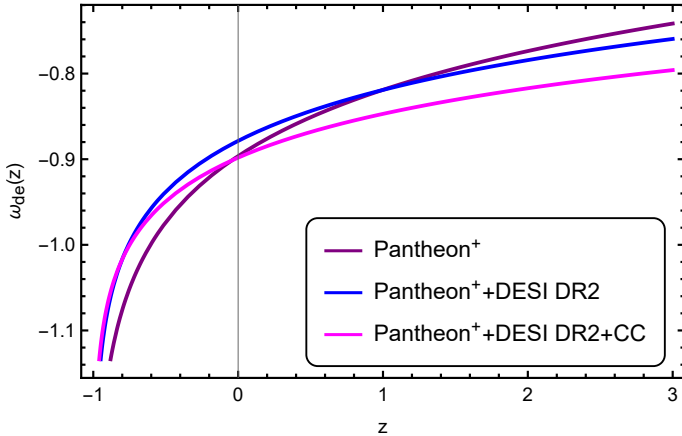


Figure 6: Evolution of the dark energy equation of state ω_{de} as a function of redshift z .

E. Density parameters

The evolution of the matter and dark energy density parameters, $\Omega_m(z)$ and $\Omega_{de}(z)$, is illustrated in Figs. 7 and 8 for the different observational datasets. As expected, $\Omega_m(z)$ increases with redshift, indicating that matter dominated the Universe in the recent past, while $\Omega_{de}(z)$ decreases with redshift, showing that dark energy only becomes dominant at late times. This complementary behavior reflects a smooth transition from a matter-dominated era to a dark energy-dominated phase driving the current accelerated expansion.

At the present epoch, the matter density parameter takes the values $\Omega_{m0} = 0.279$, 0.290 , and 0.288 for Pantheon⁺, Pantheon⁺+DESI DR2, and Pantheon⁺+DESI DR2+CC, respectively, while the corresponding dark energy density parameter values are $\Omega_{de0} = 0.720$, 0.708 , and 0.717 . These results confirm that dark energy is the dominant component at low redshift, with matter contributing a subdominant but non negligible fraction of the total energy density.

In addition, the matter density parameter $\Omega_m(z)$ is insensitive to the observational datasets. In contrast, the dark energy density parameter $\Omega_{de}(z)$ exhibits a slightly higher level of sensitivity to the choice of dataset, reflecting the mild dependence of the dark energy sector on observational constraints. Nevertheless, these variations remain small and do not affect the overall evolutionary behavior.

Overall, despite minor differences induced by the inclusion of additional observational data, the evolution of both components remains consistent, with dark energy expected to continue dominating the cosmic energy budget in the near future according to the present model.

In terms of the Hubble parameter, the density parameters are defined as

$$\Omega_m(z) = \frac{\rho_{m0} (1+z)^3}{3H^2(z)}, \quad (6.4)$$

and

$$\Omega_{de}(z) = \frac{\rho_{de0} (1+z)^{3(1+\omega_{de})}}{3H^2(z)}, \quad (6.5)$$

where ρ_{m0} and ρ_{de0} denote the present day matter and dark energy densities, respectively, and ω_{de} represents the effective dark energy equation of state.

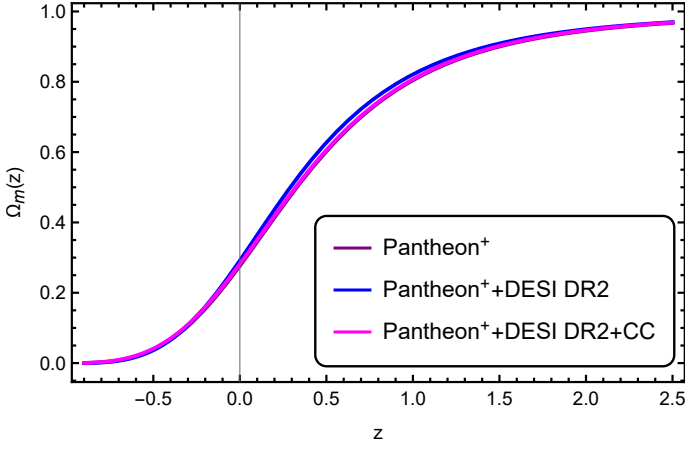


Figure 7: Evolution of the matter energy density parameter Ω_m as a function of redshift z .

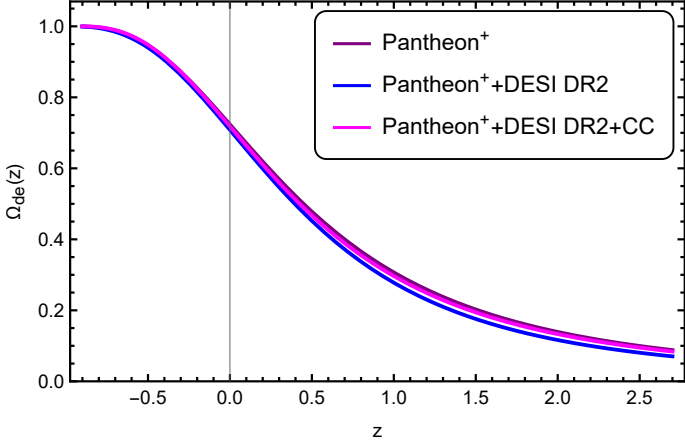


Figure 8: Evolution of the dark energy density parameter Ω_{de} as a function of redshift z .

F. Om diagnostic

The $\text{Om}(z)$ diagnostic provides a simple and robust test to distinguish the standard ΛCDM model from scenarios with dynamical dark energy, as it depends only on the Hubble parameter and is less sensitive to observational uncertainties. For a spatially flat Universe, it is defined as

$$\text{Om}(z) = \frac{E^2(z) - 1}{(1+z)^3 - 1}. \quad (6.6)$$

As illustrated in Fig. 9, the reconstructed $\text{Om}(z)$ curves for the Pantheon+, Pantheon++DESI DR2, and Pantheon++DESI DR2+CC datasets lie below the constant ΛCDM value, indicating a deviation from a strictly constant dark energy component and suggesting an effective phantom-like behavior. Moreover, the negative slope of the $\text{Om}(z)$ curves provides additional evidence for a phantom regime. The combination of multiple datasets,

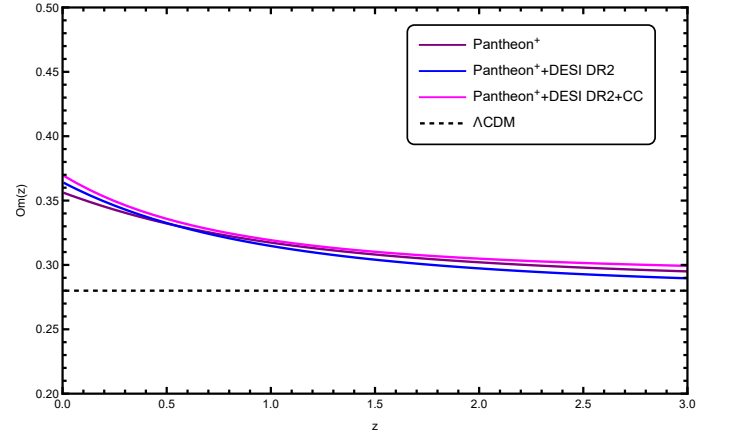


Figure 9: Evolution of the $\text{Om}(z)$ Diagnostic as a function of redshift z .

namely Pantheon++DESI DR2+CC, leads to tighter constraints and a smoother evolution. Overall, the $\text{Om}(z)$ diagnostic supports the presence of a dynamical dark energy component within the framework of power law $f(Q)$ gravity.

G. Statefinder diagnostic

The statefinder diagnostic offers a geometrical approach for discriminating among different dark energy models by involving higher order derivatives of the scale factor. It is characterized by a pair of dimensionless parameters $\{j, s\}$, which provide a model independent description of the cosmic expansion history and allow for a direct comparison between the ΛCDM model and its extensions. The statefinder parameters are defined as [80, 81]

$$s = \frac{j - 1}{3 \left(q - \frac{1}{2} \right)}, \quad (6.7)$$

where j is the jerk parameter and q denotes the deceleration parameter.

Fig. 10 displays the evolutionary trajectories in the $\{s, j\}$ plane reconstructed for the power law $f(Q)$ model using the Pantheon+, Pantheon++DESI DR2, and Pantheon++DESI DR2+CC datasets. The fixed point corresponding to the ΛCDM model, $(s = 0, j = 1)$, is marked by a star, while the solid circles indicate the present day values of the statefinder parameters for each dataset.

As shown in the figure, for all datasets, the trajectories originate in the phantom region of the $\{s, j\}$ plane and evolve toward the vicinity of the ΛCDM fixed point in the past. This transition occurs through the quintessence region, followed by a second transition toward the ΛCDM fixed point and subsequently into the phantom region in the future.

In the phantom region, before the ΛCDM fixed point, the trajectories corresponding to the Pantheon+ and Pantheon++DESI DR2 datasets coincide. In the

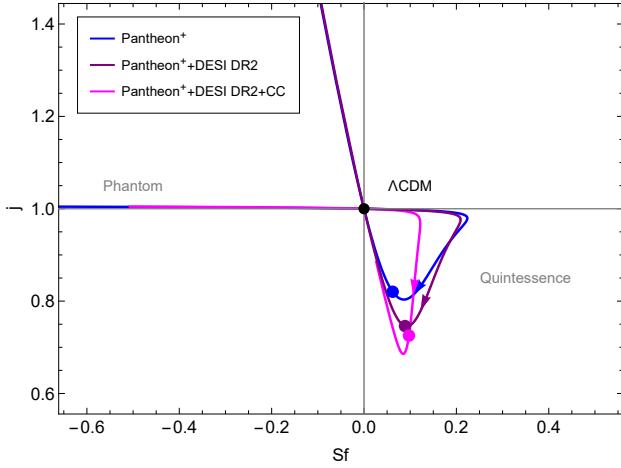


Figure 10: The evolutionary trajectories in the $\{s, j\}$ plane corresponding to the Pantheon⁺, Pantheon⁺+DESI DR2, and Pantheon⁺+DESI DR2+CC datasets. The statefinder of the Λ CDM model appears as a fixed point, marked by a star, while the solid circles indicate the present day values of the statefinder parameters.

quintessence region, the trajectories evolve differently for all dataset combinations, while after the Λ CDM fixed point, they follow a similar evolution.

VII. CONCLUSION

In this work, we investigated a cosmological scenario within the framework of symmetric teleparallel $f(Q)$ gravity, considering a power law dependence on the non metricity scalar, $f(Q) = Q + 6\gamma H_0^2 \left(\frac{Q}{Q_0}\right)^n$. The effective dark energy sector induced by non metricity was modeled through a logarithmic redshift dependent equation of state, $\omega_{de}(z) = \omega_0 + \omega_1 \ln(1+z)$, allowing for a mild evolution from a cosmological constant like behavior at low redshifts.

Using recent observational datasets including Pantheon⁺, DESI DR2, and Cosmic Chronometers we constrained model parameters via a Markov Chain Monte Carlo analysis. The results indicate that the Hubble parameter, h , and matter density, Ω_{m0} , are tightly constrained and are in close agreement with Λ CDM predictions. The dark energy equation of state remains close to $\omega_0 \simeq -1$, while the parameter ω_1 is generally small and may take either positive or negative values depending on the dataset combination, indicating a possible mild deviation from Λ CDM within observational uncertainties.

Cosmographic diagnostics of the deceleration (q), jerk (j), and snap (s) parameters reveal a smooth cosmic evolution: the Universe transitions from a past matter dominated decelerated phase to the present accelerated expansion, with higher order derivatives well behaved across all datasets. The present day values, $q_0 \simeq -0.45$, $j_0 \simeq 0.75$, and $s_0 \simeq -0.49$ (averaged over datasets), indicate that

the model closely reproduces the kinematic features of Λ CDM, while allowing for controlled deviations at earlier epochs.

The reconstructed energy density parameters, $\Omega_m(z)$ and $\Omega_{de}(z)$, confirm a smooth transition from matter dominance to dark energy dominance at low redshift, consistent with the observed accelerated expansion. The $\text{Om}(z)$ diagnostic further supports the presence of a dynamical dark energy component, showing a mild redshift dependence below the constant Λ CDM value, particularly when combining all datasets.

The analysis of the statefinder trajectories in the $\{s, j\}$ plane for the power law $f(Q)$ model shows a consistent behavior across different datasets. The trajectories originate in the phantom region, pass through the quintessence regime, and approach the Λ CDM fixed point ($s = 0, j = 1$) before evolving back into the phantom region in the future. While the paths coincide for Pantheon⁺ and Pantheon⁺+DESI DR2 datasets in the initial phantom region, the quintessence phase exhibits dataset dependent evolution, with all trajectories converging again after the Λ CDM point. This behavior highlights the model's ability to reproduce both phantom and quintessence like dynamics while remaining compatible with Λ CDM at present.

The statistical analysis using AIC_c and BIC indicates that the power law $f(Q)$ model provides a competitive description of the observational data. For Pantheon⁺+DESI DR2, a moderate preference is found according to AIC_c , while the inclusion of CC slightly weakens this preference but tightens the parameter constraints, bringing them closer to the Λ CDM limit. Although BIC penalizes the model due to its additional free parameters, the overall results demonstrate that the power law $f(Q)$ scenario remains statistically consistent and phenomenologically viable, allowing for controlled deviations from a constant dark energy equation of state while staying compatible with Λ CDM.

Overall, our analysis demonstrates that the power law $f(Q)$ model offers a consistent and phenomenologically viable description of the late time Universe. It provides a geometric alternative to dark energy, capable of reproducing the observed expansion history, and highlights the role of non metricity in shaping cosmic dynamics. These results motivate further theoretical and observational studies to explore extensions of the model and the fundamental geometric properties of gravity beyond Riemannian frameworks.

Future perspectives include the use of additional observational probes, such as redshift space distortions, and the cosmic microwave background, to test the model more rigorously and to identify potential deviations from Λ CDM, particularly at the level of perturbations. From a theoretical point of view, it is valuable to explore extensions of the $f(Q)$ model, consider alternative parameterizations of the dark energy equation of state, and analyze the evolution of cosmological perturbations and model stability to assess its robustness beyond the background evolution. These directions provide promising avenues to more clearly distinguish modified gravity scenarios from the standard cosmo-

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- [1] Perlmutter, S., Aldering, G., Goldhaber, G., Knop, R. A., Nugent, P., Castro, P. G., *et al.* & Supernova Cosmology Project, *Measurements of Ω and Λ from 42 high-redshift supernovae*, *The Astrophysical Journal*, **517**(2), 565 (1999).
- [2] Riess, A. G., Filippenko, A. V., Challis, P., Clocchiatti, A., Diercks, A., Garnavich, P. M., *et al.* & Tonry, J., *Observational evidence from supernovae for an accelerating universe and a cosmological constant*, *The Astronomical Journal*, **116**(3), 1009 (1998).
- [3] Peebles, P. J. E., & Ratra, B., *The cosmological constant and dark energy*, *Reviews of Modern Physics*, **75**(2), 559 (2003).
- [4] Padmanabhan, T., *Cosmological constant—the weight of the vacuum*, *Physics Reports*, **380**(5-6), 235-320 (2003).
- [5] Copeland, E. J., Sami, M., & Tsujikawa, S., *Dynamics of dark energy*, *International Journal of Modern Physics D*, **15**(11), 1753-1935 (2006).
- [6] Frieman, J. A., Turner, M. S., & Huterer, D., *Dark energy and the accelerating universe*, *Annual Review of Astronomy and Astrophysics*, **46**(1), 385-432 (2008).
- [7] Caldwell, R. R., *A phantom menace? Cosmological consequences of a dark energy component with super-negative equation of state*, *Physics Letters B*, **545**(1-2), 23-29 (2002).
- [8] Silvestri, A., & Trodden, M., *Approaches to modifying gravity for dark energy*, *Reports on Progress in Physics*, **72**, 096901 (2009).
- [9] Bamba, K., & Capozziello, S., Nojiri, S., & Odintsov, S. D., *Dark energy cosmology: the equivalent description via different theoretical models and cosmography tests*, *Astrophysics and Space Science*, **342**, 155 (2012).
- [10] Li, M., Li, X. D., Wang, S., & Wang, Y., *Dark energy*, *Communications in Theoretical Physics*, **56**(3), 525 (2011).
- [11] Carroll, S. M., *Quintessence and the rest of the world*, arXiv:astro-ph/9806099 (1998).
- [12] Ladghami, Y., Asfour, B., Bargach, A., Errahmani, A., & Ouali, T., *Thermodynamics of 4D-EGB black holes in the quintessential phase space*, *Annals of Physics*, **460**, 169575 (2024).
- [13] Bouali, A., Albarran, I., Bouhmadi-López, M., Errahmani, A., & Ouali, T., *Cosmological constraints of interacting phantom dark energy models*, *Physics of the Dark Universe*, **34**, 100907 (2021).
- [14] Bouali, A., Albarran, I., Bouhmadi-López, M., & Ouali, T., *Cosmological constraints of phantom dark energy models*, *Physics of the Dark Universe*, **26**, 100391 (2019).
- [15] Mhamdi, D., Bargach, F., Dahmani, S., Bouali, A., & Ouali, T., *Comparing phantom dark energy models with various diagnostic tools*, *General Relativity and Gravitation*, **55**(1), 11 (2023).
- [16] Dahmani, S., Bouali, A., El Bojaddaini, I., Errahmani, A., & Ouali, T., *Constraining neutrino properties and smoothing the Hubble tension via the LSBR model*, *General Relativity and Gravitation*, **55**(1), 22 (2023).
- [17] Dahmani, S., Bouali, A., El Bojaddaini, I., Errahmani, A., & Ouali, T., *Smoothing the H_0 tension with a phantom dynamical dark energy model*, *Physics of the Dark Universe*, **42**, 101266 (2023).
- [18] Bouhmadi-Lopez, M., Errahmani, A., Martin-Moruno, P., Ouali, T., & Tavakoli, Y., *The little sibling of the big rip singularity*, *International Journal of Modern Physics D*, **24**(10), 1550078 (2015).
- [19] Malquarti, M., Copeland, E. J., Liddle, A. R., & Trodden, M., *A new view of k-essence*, *Physical Review D*, **67**(12), 123503 (2003).
- [20] Chiba, T., Okabe, T., & Yamaguchi, M., *Kinetically driven quintessence*, *Physical Review D*, **62**(2), 023511 (2000).
- [21] Horava, P., & Minic, D., *Probable values of the cosmological constant in a holographic theory*, *Physical Review Letters*, **85**(8), 1610 (2000).
- [22] Li, M., *A model of holographic dark energy*, *Physics Letters B*, **603**(1-2), 1-5 (2004).
- [23] Wang, S., Wang, Y., & Li, M., *Holographic dark energy*, *Physics Reports*, **696**, 1-57 (2017).
- [24] Bouhmadi-López, M., Errahmani, A., Ouali, T., & Tavakoli, Y., *More on the holographic Ricci dark energy model: smoothing Rips through interaction effects?*, *The European Physical Journal C*, **78**(4), 330 (2018).
- [25] Belkacemi, M. H., Bouhmadi-López, M., Errahmani, A., & Ouali, T., *Holographic induced gravity model with a Ricci dark energy: Smoothing the little rip and big rip through Gauss-Bonnet effects?*, *Physical Review D — Particles, Fields, Gravitation, and Cosmology*, **85**(8), 083503 (2012).
- [26] Bargach, F., Bargach, A., Bouhmadi-López, M., Errahmani, A., & Ouali, T., *A dynamical system approach to the holographic dark energy in a modified theory of gravity*, *International Journal of Modern Physics D*, **30**(10), 2150076 (2021).
- [27] Bouhmadi-López, M., Errahmani, A., & Ouali, T., *Cosmology of a holographic induced gravity model with curvature effects*, *Physical Review D — Particles, Fields, Gravitation, and Cosmology*, **84**(8), 083508 (2011).
- [28] Belkacemi, M. H., Bouabdallaoui, Z., Bouhmadi-López, M., Errahmani, A., & Ouali, T., *An interacting holographic dark energy model within an induced gravity brane*, *International Journal of Modern Physics D*, **29**(09), 2050066 (2020).
- [29] Capozziello, S., *Curvature quintessence*, *International Journal of Modern Physics D*, **11**(04), 483-491 (2002).
- [30] Nojiri, S. I., & Odintsov, S. D., *Modified $f(R)$ gravity consistent with realistic cosmology: From a matter dominated epoch to a dark energy universe*, *Physical Review D — Particles, Fields, Gravitation, and Cosmology*, **74**(8), 086005 (2006).
- [31] Appleby, S. A., & Battye, R. A., *Do consistent $F(R)$ models mimic general relativity plus Λ ?*, *Physics Letters B*, **654**(1-2), 7-12 (2007).
- [32] Hu, W., & Sawicki, I., *Models of $f(R)$ cosmic acceleration that evade solar system tests*, *Physical Review D — Particles, Fields, Gravitation, and Cosmology*, **76**(6), 064004 (2007).
- [33] Starobinsky, A. A., *Disappearing cosmological constant in $f(R)$ gravity*, *JETP Letters*, **86**(3), 157-163 (2007).
- [34] Bertolami, O., Boehmer, C. G., Harko, T., & Lobo, F. S., *Extra force in $f(R)$ modified theories of gravity*, *Physical Review D — Particles, Fields, Gravitation, and Cosmology*, **75**(10), 104016 (2007).

- [35] Harko, T., & Lobo, F. S., $f(R, L_m)$ gravity, *The European Physical Journal C*, **70**(1-2), 373-379 (2010).
- [36] Harko, T., Lobo, F. S., Nojiri, S. I., & Odintsov, S. D., $f(R, T)$ gravity, *Physical Review D — Particles, Fields, Gravitation, and Cosmology*, **84**(2), 024020 (2011).
- [37] Haghani, Z., Harko, T., Sepangi, H. R., & Shahidi, S., Weyl-Cartan-Weitzenböck gravity as a generalization of teleparallel gravity, *Journal of Cosmology and Astroparticle Physics*, **2012**(10), 061 (2012).
- [38] Harko, T., & Shahidi, S., Coupling matter and curvature in Weyl geometry: conformally invariant $f(R, L_m)$ gravity, *The European Physical Journal C*, **82**(3), 219 (2022).
- [39] Harko, T., & Shahidi, S., Palatini formulation of the conformally invariant $f(R, L_m)$ gravity theory, *The European Physical Journal C*, **82**(11), 1003 (2022).
- [40] Akarsu, Ö., & Kılınç, C. B., Bianchi type III models with anisotropic dark energy, *General Relativity and Gravitation*, **42**(4), 763-775 (2010).
- [41] Marra, V., Pääkkönen, M., & Valkenburg, W., Uncertainty on w from large-scale structure, *Monthly Notices of the Royal Astronomical Society*, **431**(2), 1891-1902 (2013).
- [42] Valkenburg, W., Perceiving the equation of state of Dark Energy while living in a Cold Spot, *Journal of Cosmology and Astroparticle Physics*, **2012**(01), 047 (2012).
- [43] Amendola, L., Campos, G. C., & Rosenfeld, R., Consequences of dark matter–dark energy interaction on cosmological parameters derived from type Ia supernova data, *Physical Review D — Particles, Fields, Gravitation, and Cosmology*, **75**(8), 083506 (2007).
- [44] Clemson, T., Koyama, K., Zhao, G. B., Maartens, R., & Väliiviita, J., Interacting dark energy: Constraints and degeneracies, *Physical Review D — Particles, Fields, Gravitation, and Cosmology*, **85**(4), 043007 (2012).
- [45] Lu, J., Wu, Y., Jin, Y., & Wang, Y., Investigate the interaction between dark matter and dark energy, *Results in Physics*, **2**, 14-21 (2012).
- [46] Caldwell, R. R., Dave, R., & Steinhardt, P. J., Cosmological imprint of an energy component with general equation of state, *Physical Review Letters*, **80**(8), 1582 (1998).
- [47] Sahni, V., & Habib, S., Does inflationary particle production suggest $\Omega_m < 1$?, *Physical Review Letters*, **81**(9), 1766 (1998).
- [48] Zlatev, I., Wang, L., & Steinhardt, P. J., Quintessence, cosmic coincidence, and the cosmological constant, *Physical Review Letters*, **82**(5), 896 (1999).
- [49] Matarrese, S., Baccigalupi, C., & Perrotta, F., Approaching Λ without fine-tuning, *Physical Review D — Particles, Fields, Gravitation, and Cosmology*, **70**(6), 061301 (2004).
- [50] Pettorino, V., Baccigalupi, C., & Perrotta, F., Scaling solutions in scalar–tensor cosmologies, *Journal of Cosmology and Astroparticle Physics*, **2005**(12), 003 (2005).
- [51] Gupta, G., Saridakis, E. N., & Sen, A. A., Nonminimal quintessence and phantom with nearly flat potentials, *Physical Review D — Particles, Fields, Gravitation, and Cosmology*, **79**(12), 123013 (2009).
- [52] Cervantes-Cota, J. L., de Putter, R., & Linder, E. V., Induced gravity and the attractor dynamics of dark energy/dark matter, *Journal of Cosmology and Astroparticle Physics*, **2010**(12), 019 (2010).
- [53] de la Macorra, A., Scalar field dark energy parametrization, arXiv:1108.0876 [astro-ph.CO] (2011).
- [54] Starobinsky, A. A., A new type of isotropic cosmological models without singularity, *Physics Letters B*, **91**(1), 99-102 (1980).
- [55] De Felice, A., & Tsujikawa, S., Construction of cosmologically viable $f(G)$ dark energy models, *Physics Letters B*, **675**(1), 1 (2009).
- [56] Erices, C., Papantonopoulos, E., & Saridakis, E. N., Cosmology in cubic and $f(P)$ gravity, *Physical Review D*, **99**(12), 123527 (2019).
- [57] Deffayet, C., Esposito-Farese, G., & Vikman, A., Covariant galileon, *Physical Review D — Particles, Fields, Gravitation, and Cosmology*, **79**(8), 084003 (2009).
- [58] Cai, Y. F., Capozziello, S., De Laurentis, M., & Saridakis, E. N., $f(T)$ teleparallel gravity and cosmology, *Reports on Progress in Physics*, **79**(10), 106901 (2016).
- [59] Jiménez, J. B., Heisenberg, L., Koivisto, T., & Pekar, S., Cosmology in $f(Q)$ geometry, *Physical Review D*, **101**(10), 103507 (2020).
- [60] Nester, J. M., & Yo, H.-J., Symmetric teleparallel general relativity, *Chinese Journal of Physics*, **37**, 113 (1999).
- [61] Jiménez, J. B., Heisenberg, L., & Koivisto, T., Coincident general relativity, *Physical Review D*, **98**(4), 044048 (2018).
- [62] Albuquerque, I. S., & Frusciante, N., A designer approach to $f(Q)$ gravity and cosmological implications, *Physics of the Dark Universe*, **35**, 100980 (2022).
- [63] Solanki, R., De, A., & Sahoo, P. K., Complete dark energy scenario in $f(Q)$ gravity, *Physics of the Dark Universe*, **36**, 100996 (2022).
- [64] Xu, Y., Li, G., Harko, T., & Liang, S. D., $f(Q, T)$ gravity, *The European Physical Journal C*, **79**(8), 708 (2019).
- [65] Efstathiou, G., Constraining the equation of state of the universe from distant type Ia supernovae and cosmic microwave background anisotropies, *Monthly Notices of the Royal Astronomical Society*, **310**(3), 842-850 (1999).
- [66] Mehrabi, A., Basilakos, S., & Pace, F., How clustering dark energy affects matter perturbations, *Monthly Notices of the Royal Astronomical Society*, **452**(3), 2930-2939 (2015).
- [67] Mandal, S., Sokoliuk, O., Mishra, S. S., & Sahoo, P. K., H_0 tension in torsion-based modified gravity, *Nuclear Physics B*, **993**, 116285 (2023).
- [68] Scolnic, D., Brout, D., Carr, A., Riess, A. G., Davis, T. M., Dwomoh, A., et al., The Pantheon+ analysis: the full data set and light-curve release, *The Astrophysical Journal*, **938**(2), 113 (2022).
- [69] J. Niu, P. He, and T. J. Zhang, Constraining the Hubble constant with a simulated full covariance matrix using neural networks, *Monthly Notices of the Royal Astronomical Society*, vol. 545, no. 1, staf2051, 2026.
- [70] Moresco, R., Jimenez, L., Verde, A., Cimatti, and L. Pozzetti, Setting the stage for cosmic chronometers. II. Impact of stellar population synthesis models systematics and full covariance matrix, *The Astrophysical Journal*, vol. 898, no. 1, p. 82, 2020.
- [71] Akaike, H., A new look at the statistical model identification, *IEEE Transactions on Automatic Control*, **19**(6), 716-723 (2003).
- [72] Vrieze, S. I., Model selection and psychological theory: a discussion of the differences between the Akaike information criterion (AIC) and the Bayesian information criterion (BIC), *Psychological Methods*, **17**(2), 228 (2012).
- [73] Lewis, A., & Bridle, S., Cosmological parameters from CMB and other data: A Monte Carlo approach, *Physical Review D*, **66**(10), 103511 (2002).
- [74] Gilks, W. R., Richardson, S., & Spiegelhalter, D. (Eds.), *Markov Chain Monte Carlo in Practice*, CRC Press (1995).
- [75] Mhamdi, D., Bargach, F., Dahmani, S., Bouali, A., & Ouali, T. (2023). Comparing phantom dark energy models with

- various diagnostic tools*. General Relativity and Gravitation, 55(1), 11.
- [76] Bouali, A., Chaudhary, H., Harko, T., Lobo, F. S., Ouali, T., & Pinto, M. A. (2023). *Observational constraints and cosmological implications of scalar-tensor $f(R, T)$ gravity*. Monthly Notices of the Royal Astronomical Society, 526(3), 4192-4208.
 - [77] Mandal, S., Pradhan, S., Sahoo, P. K., & Harko, T. (2023). Cosmological observational constraints on the power law $f(Q)$ type modified gravity theory. *The European Physical Journal C*, 83(12), 1141.
 - [78] Ayuso, I., Lazkoz, R., & Salzano, V. (2021). Observational constraints on cosmological solutions of $f(Q)$ theories. *Physical Review D*, 103(6), 063505.
 - [79] Abdul Karim, M., Aguilar, J., Ahlen, S., Alam, S., Allen, L., Prieto, C. A., *et al.* (2025). DESI DR2 results. II. Measurements of baryon acoustic oscillations and cosmological constraints. *Physical Review D*, 112(8), 083515.
 - [80] Sahni, V., Saini, T. D., Starobinsky, A. A., & Alam, U., *Statefinder—a new geometrical diagnostic of dark energy*, *Journal of Experimental and Theoretical Physics Letters*, 77(5), 201-206 (2003).
 - [81] Alam, U., Sahni, V., Deep Saini, T., & Starobinsky, A. A., *Exploring the expanding universe and dark energy using the Statefinder diagnostic*, *Monthly Notices of the Royal Astronomical Society*, 344(4), 1057-1074 (2003).
 - [82] Jiménez, J. B., Heisenberg, L., & Koivisto, T., *Coincident general relativity*, *Physical Review D*, 98(4), 044048 (2018).
 - [83] Moresco, M., Jimenez, R., Verde, L., Cimatti, A., & Pozzetti, L., *Setting the stage for cosmic chronometers. II. Impact of stellar population synthesis models systematics and full covariance matrix*, *The Astrophysical Journal*, 898(1), 82 (2020).
 - [84] Mandal, S., Pradhan, S., Sahoo, P. K., & Harko, T., *Cosmological observational constraints on the power law $f(Q)$ type modified gravity theory*, *The European Physical Journal C*, 83(12), 1141 (2023).
 - [85] Koussour, M., El Bourakadi, K., Shekh, S. H., Pacif, S. K. J., & Bennai, M., *Late-time acceleration in $f(Q)$ gravity: Analysis and constraints in an anisotropic background*, *Annals of Physics*, 445, 169092 (2022).
 - [86] Koussour, M., Pacif, S. K. J., Bennai, M., & Sahoo, P. K., *A new parametrization of Hubble parameter in $f(Q)$ gravity*, *Fortschritte der Physik*, 71(4-5), 2200172 (2023).
 - [87] Mandal, S., Pradhan, S., Sahoo, P. K., & Harko, T., *Cosmological observational constraints on the power law $f(Q)$ type modified gravity theory*, *The European Physical Journal C*, 83(12), 1141 (2023).
 - [88] Koussour, M., El Bourakadi, K., Shekh, S. H., Pacif, S. K. J., and Bennai, M. (2022). Late-time acceleration in $f(Q)$ gravity: Analysis and constraints in an anisotropic background. *Annals of Physics*, 445, 169092.
 - [89] Abdul Karim, M., Aguilar, J., Ahlen, S., Alam, S., Allen, L., Prieto, C. A., *et al.* (2025). DESI DR2 results. II. Measurements of baryon acoustic oscillations and cosmological constraints. *Physical Review D*, 112(8), 083515.